

Maths - Question 57

Original paper page 46; original question and option regions



If the direction cosines of two lines are given by $l + m + n = 0$ and $mn - 2lm - 2nl = 0$, then the acute angle between those lines is

రెండు రేఖల దిక్ కొసైన్లు $l + m + n = 0$ మరియు $mn - 2lm - 2nl = 0$ గా ఇచ్చినప్పుడు ఆ రేఖల మధ్యనుండే అల్పకోణం

Options :

1. ✖ $2\pi/5$

2. ✔ $\pi/3$

3. ✖ $\pi/4$

4. ✖ $\pi/60$

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Maths - Question 58

Original paper page 46; original question and option regions

If the angle θ between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and the plane $2x - y + \sqrt{\lambda} z + 4 = 0$ is such that $\sin \theta = \frac{1}{3}$, then the value of $\lambda =$

$\sin \theta = \frac{1}{3}$ అయ్యేటట్లుగా రేఖ $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ మరియు తలము $2x - y + \sqrt{\lambda} z + 4 = 0$

మధ్య గల కోణము θ ఉంటే, అప్పుడు λ యొక్క విలువ

Maths - Question 58 (continued)

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$$\frac{3}{5}$$

2. ✖

$$\frac{5}{4}$$

3. ✔

$$\frac{5}{3}$$

4. ✖

$$\frac{4}{3}$$



Maths - Question 59

Original paper page 47; original question and option regions

Let $f(x) = \begin{cases} 1 + \frac{2x}{a}, & 0 \leq x \leq 1 \\ ax, & 1 < x \leq 2 \end{cases}$. If $\lim_{x \rightarrow 1} f(x)$ exists then the sum of the cubes of the possible values of a is

$f(x) = \begin{cases} 1 + \frac{2x}{a}, & 0 \leq x \leq 1 \\ ax, & 1 < x \leq 2 \end{cases}$ అనుకోండి. $\lim_{x \rightarrow 1} f(x)$ వ్యవస్థితం అయితే, a కు సాధ్యమయ్యే విలువల ఘనాల మొత్తం

Options :

1. ✖ 1

2. ✖ 5

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Maths - Question 60

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Let $[P]$ denote the greatest integer $\leq P$. If $0 \leq a \leq 2$, then the number of integral values of 'a' such that $\lim_{x \rightarrow a} ([x^2] - [x]^2)$ does not exist is

P కి మించని గరిష్ఠ పూర్ణాంకాన్ని $[P]$ సూచిస్తుందనుకోండి. $0 \leq a \leq 2$ అయితే,

$\lim_{x \rightarrow a} ([x^2] - [x]^2)$ వ్యవస్థితం కాకుండా ఉండేటట్లు ఉన్న 'a' యొక్క పూర్ణాంక విలువల

సంఖ్య

Options :

1. ✖ 3

2. ✔ 2

3. ✖ 1

4. ✖ 0

Maths - Question 61

Original paper page 49; original question and option regions

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A 

Time : 0

If $f(x) = \begin{cases} \frac{\sqrt{a^2 - ax + x^2} - \sqrt{x^2 + ax + a^2}}{\sqrt{a+x} - \sqrt{a-x}}, & x \neq 0 \\ K, & x = 0 \end{cases}$ is continuous at $x = 0$, then $K =$

$x = 0$ వద్ద $f(x) = \begin{cases} \frac{\sqrt{a^2 - ax + x^2} - \sqrt{x^2 + ax + a^2}}{\sqrt{a+x} - \sqrt{a-x}}, & x \neq 0 \\ K, & x = 0 \end{cases}$, అవిచ్ఛిన్నము అయితే, $K =$

Options :

1. ✓ $-\sqrt{a}$

2. ✗ $\sqrt{a}$

3. ✗ -1

4. ✗ $a + \sqrt{a}$

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Maths - Question 62

Original paper page 49; original question and option regions

$$\text{If } y = \sinh^{-1} \left(\frac{1-x}{1+x} \right) \text{ then } \frac{dy}{dx} =$$

$$y = \sinh^{-1} \left(\frac{1-x}{1+x} \right) \implies \frac{dy}{dx} =$$

Maths - Question 62 (continued)

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$$\frac{-\sqrt{2}}{|1+x| \sqrt{1+x^2}}$$

2. ✖

$$\frac{-1}{(1+x) \sqrt{x}}$$

3. ✖

$$\frac{1}{(1+x^2) \sqrt{1+x}}$$

4. ✖

$$\frac{-\sqrt{2}}{(1+x) \sqrt{1-x}}$$

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Maths - Question 63

Original paper page 50; original question and option regions

If $y = (x - 1)(x + 2)(x^2 + 5)(x^4 + 8)$, then $\lim_{x \rightarrow -1} \left(\frac{dy}{dx} \right) =$

$y = (x - 1)(x + 2)(x^2 + 5)(x^4 + 8)$ எனில், $\lim_{x \rightarrow -1} \left(\frac{dy}{dx} \right) =$

Options :

-30

1. ✖

30

2. ✔

Maths - Question 63 (continued)

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Maths - Question 64

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If $f(x) = \begin{cases} ax^2 + bx - \frac{13}{8}, & x \leq 1 \\ 3x - 3, & 1 < x \leq 2 \\ bx^3 + 1, & x > 2 \end{cases}$ is differentiable $\forall x \in R$, then $a - b =$

అన్ని $x \in R$ లకు $f(x) = \begin{cases} ax^2 + bx - \frac{13}{8}, & x \leq 1 \\ 3x - 3, & 1 < x \leq 2 \\ bx^3 + 1, & x > 2 \end{cases}$ అవకలనీయము అయితే, $a - b =$

Options :

1. ✓ $\frac{9}{8}$

2. ✗ $\frac{5}{4}$

3. ✗ $\frac{11}{8}$

4. ✗ $\frac{1}{4}$

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Maths - Question 64 (continued)

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A is a point on the circle with radius 8 and centre at O. A particle P is moving on the circumference of the circle starting from A. M is the foot of the perpendicular from P on OA and $\angle POM = \theta$. When $OM=4$ and $\frac{d\theta}{dt} = 6$ radians/sec, then the rate of change of PM is (in units/sec)

O వద్ద కేంద్రం మరియు 8 వ్యాసార్థం గా గల ఒక వృత్తం పై A ఒక బిందువు. A నుండి బయలుదేరి ఒక కణము P వృత్త పరిధి పై కదులుతోంది. P నుండి OA కు గీచిన లంబపాదం M మరియు $\angle POM = \theta$. $OM = 4$ మరియు $\frac{d\theta}{dt} = 6$ రేడియన్స్ / సెకండ్ అయినప్పుడు PM లోని మార్పు రేటు (సెకనుకు యూనిట్లలో)

Options :

1. ✖ $24\sqrt{3}$

2. ✔ 24

3. ✖ $15\sqrt{3}$

4. ✖ $48\sqrt{3}$

The function $f(x) = \begin{cases} \frac{x-|x|}{x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$

$f(x) = \begin{cases} \frac{x-|x|}{x}, & x \neq 0 \\ 2, & x = 0 \end{cases}$ అనే ప్రమేయము

Options :

is continuous for $\forall x \in R$

ప్రతి $x \in R$ వద్ద అవిచ్ఛిన్నము

1. ✖

has maximum value 2

గరిష్ట విలువ 2 ను కలిగిఉంటుంది

2. ✔

Maths - Question 68 (continued)

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has neither minimum nor maximum

కనిష్ఠ విలువ మరియు గరిష్ఠ విలువ ఉండవు

has minimum value 2

కనిష్ఠ విలువ 2 ను కలిగిఉంటుంది

Maths - Question 69

Original paper page 55; original question and option regions

If $\int \frac{\sqrt[4]{x}}{\sqrt{x} + \sqrt[4]{x}} dx = \frac{2}{3} [A \sqrt[4]{x^3} + B \sqrt[4]{x^2} + C \sqrt[4]{x} + D \log(1 + \sqrt[4]{x})] + K$ then $\frac{2}{3} (A+B+C+D) =$

$\int \frac{\sqrt[4]{x}}{\sqrt{x} + \sqrt[4]{x}} dx = \frac{2}{3} [A \sqrt[4]{x^3} + B \sqrt[4]{x^2} + C \sqrt[4]{x} + D \log(1 + \sqrt[4]{x})] + K$ అయితే $\frac{2}{3} (A+B+C+D) =$

Options :

1. ✖

$$\frac{2}{3}$$

2. ✔

$$-\frac{2}{3}$$

3. ✖

$$\frac{4}{3}$$

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Maths - Question 69 (continued)

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$$-\frac{4}{3}$$



Maths - Question 70

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$$\int (\log x)^m x^n dx =$$

Options :

$$\int t^m e^{nt} dt, \quad t = e^x$$

1. ✖

$$\int t^m e^{(n+1)t} dt, \quad t = e^x$$

2. ✖

$$\int t^m e^{(n+1)t} dt, \quad x = e^t$$

3. ✔

$$\int t^m e^{nt} dt, \quad x = e^t$$

4. ✖

Maths - Question 71

Original paper page 57; original question and option regions



$$\int \sin^{-1} \left(\sqrt{\frac{x-a}{x}} \right) dx =$$

Options :

1. ✓ $x \cos^{-1} \sqrt{\frac{a}{x}} - \sqrt{ax - a^2} + c$

2. ✗ $x \sec^{-1} \sqrt{\frac{a}{x}} + \sqrt{x^2 - ax} + c$

3. ✗ $x \sin^{-1} \sqrt{\frac{x}{a}} + \sqrt{x^2 + ax} + c$

4. ✗ $\frac{x}{a} \sin^{-1} \frac{x}{a} + \frac{x^2}{a} \sqrt{1+a^2} + c$

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Maths - Question 72

Original paper page 57; original question and option regions

If $\int \frac{\sin x \cos x}{\sqrt{\cos^4 x - \sin^4 x}} dx = -\frac{f(x)}{2} + c$, then domain of $f(x)$ is

$\int \frac{\sin x \cos x}{\sqrt{\cos^4 x - \sin^4 x}} dx = -\frac{f(x)}{2} + c$ అయితే $f(x)$ యొక్క ప్రదేశము

Options :

$$[2n\pi, (2n+1)\pi], n = 0, 1, 2, \dots$$

Maths - Question 72 (continued)

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$$\left[(4n - 1) \frac{\pi}{2}, (4n+1) \frac{\pi}{2} \right], n = 0, 1, 2, \dots$$

$$\left[(4n - 1) \frac{\pi}{4}, (4n+1) \frac{\pi}{4} \right], n = 0, 1, 2, \dots$$

$$\left[(2n \frac{\pi}{4}, (2n+1) \frac{\pi}{4} \right], n = 0, 1, 2, \dots$$



Maths - Question 73

Original paper page 58; original question and option regions

If $y = (\tan^{-1} 2x)^2 + (\cot^{-1} 2x)^2$, then $(1+4x^2)^2 y'' - 16 =$

$y = (\tan^{-1} 2x)^2 + (\cot^{-1} 2x)^2$, అయితే $(1+4x^2)^2 y'' - 16 =$

Options :

1. ✖ $8x y'$

2. ✔ $-8x(1 + 4x^2)y'$

3. ✖ $8x(1 + 4x^2)y'$

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Maths - Question 73 (continued)

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$$- 8xy'$$



Maths - Question 74

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If $\int_0^{2\pi} (\sin^4 x + \cos^4 x) dx = K \int_0^\pi \sin^2 x dx + L \int_0^{\frac{\pi}{2}} \cos^2 x dx$ and $K, L \in \mathbb{N}$, then the number of possible ordered pairs (K, L) is

$\int_0^{2\pi} (\sin^4 x + \cos^4 x) dx = K \int_0^\pi \sin^2 x dx + L \int_0^{\frac{\pi}{2}} \cos^2 x dx$ మరియు $K, L \in \mathbb{N}$ అయిన సాధ్యమయ్యే (K, L) క్రమ యుగ్మాల సంఖ్య

Options :

1. ✖

2. ✔

3. ✖

4. ✖

Maths - Question 75

Original paper page 59; original question and option regions

$$\int_0^{\pi} \frac{x \sin x}{4\cos^2 x + 3\sin^2 x} dx =$$

Maths - Question 75 (continued)

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ons :



$$\frac{\pi^2}{6\sqrt{3}}$$

$$\frac{\pi}{3\sqrt{3}}$$

$$\frac{\pi^2}{3\sqrt{3}}$$

$$\sqrt{3} \pi^2$$

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Maths - Question 76

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If $A = \int_0^{\infty} \frac{1+x^2}{1+x^4} dx$, $B = \int_0^1 \frac{1+x^2}{1+x^4} dx$, then

$$A = \int_0^{\infty} \frac{1+x^2}{1+x^4} dx, \quad B = \int_0^1 \frac{1+x^2}{1+x^4} dx$$

Options :

$$2A = B$$

1. ✖

$$A = B$$

2. ✖

Maths - Question 76 (continued)

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$$2B = A$$

$$2B + A = 0$$



If (α, β) is the stationary point of the curve $y = 2x - x^2$, then the area bounded by the curves

$y = 2^x$, $y = 2x - x^2$, $x = 0$ and $x = \alpha$ is

$y = 2x - x^2$ వక్రానికి (α, β) విరామ బిందువైతే, వక్రాలు $y = 2^x$, $y = 2x - x^2$, $x = 0$ మరియు $x = \alpha$ లచే పరివృతమయ్యే ప్రాంత వైశాల్యం

Options :

$$\frac{3\log 2 + 4}{2}$$

1. ✖

$$\frac{3 + \log 4}{6}$$

2. ✖

$$\frac{3 - \log 4}{3}$$

3. ✔

$$\frac{1}{\log 2} + \frac{3}{4}$$

4. ✖

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Maths - Question 77 (continued)

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Maths - Question 80

Original paper page 63; original question and option regions

The solution of the differential equation $e^x y dx + e^x dy + x dx = 0$ is

$e^x y dx + e^x dy + x dx = 0$ అవకలన సమీకరణం యొక్క సాధన

Options :

$$e^x + yx^2 = c$$

1. ✖

Maths - Question 80 (continued)

Original paper page 64; original question and option regions



$$2ye^x + x^2 = c$$

3. ✖

$$ye^x + x^2 e^y = c$$

4. ✖

$$e^x + xe^y = c$$